Jewels: Toward Text-Conditioned, Streamable, and Locally Editable Video as Space-Time Gaussian Fields

The Jewels Project

Technical report – August 13, 2026

Status. This is a progress report, not a benchmark or a claim of completed text-to-video generation. It separates established results, failed controls, and hypotheses still under test.

Abstract

We investigate whether ordinary RGB video can be generated and edited as an explicit field of anisotropic Gaussian primitives in normalized image-time coordinates rather than as a sequence of pixel frames. Each primitive, or *jewel*, occupies a volume in (u,v,t), carries opacity and a linear color field, and may tilt through time so that motion is represented by oriented space-time support. A deterministic additive renderer turns the set into frames. The longer-term model emits new jewels in overlapping time windows, retains persistent jewels by stable identity, and repairs only dirty latent cells after a user moves selected primitives.

This report consolidates the feasibility evidence to date. Synthetic tubes reconstruct at about 55 dB; four finalized 96-frame UCF-101 fields with 120k jewels replay at a mean 32.28 dB. A density audit shows that raw jewel count is misleading: doubling a UCF field from 45k to 90k isotropically increases effective per-frame contributors by only 30.3% and lowers full-volume PSNR by 1.07 dB, motivating temporal-preserving spatial densification. A local $32^3 \times 24$ tokenizer is the first held-out configuration to preserve small pedestrians; a train-only 2^3 PCA hierarchy reduces it to $16^3 \times 96$ with a 0.224 dB penalty. An axial rectified-flow prior beats retrieval and mean-collapse distribution baselines but does not use its semantic condition. Conversely, a cell-local prefix model reduces future-birth feature error by 62.7% against a disjoint shuffled-prefix control and renders at 19.87 dB versus 7.30 dB. Edit planning and masked sampling preserve every clean latent exactly, although the repaired region remains blurry. A four-class direct prompt model changes held-out birth statistics but produces washed-out action geometry. Exact target topology improves its mean field PSNR by only 0.172 dB, rejecting count prediction as the primary cause. Oracle-topology stochastic flow recovers edge energy without coherent objects; adding the true future as a 24×40 guide raises held-out rendering from 14.232 to 16.555 dB and from 59.8% to 90.5% of target edge energy. This privileged control selects a semantic-video-scaffold-to-jewel architecture; it is not a prompt-only generation result. Official distilled LTX-2.3 then produces all 16 balanced prompt scaffolds. Four density-matched 49-frame/72k-jewel reconstructions replay at 28.18 dB to 35.16 dB, averaging 5,966 effective contributors/frame. Token-only multiscale, raster-plus-token, and two raster-guide render-loss controls fail to dominate the original guide. The selected UCF-trained realizer then passes a predeclared cross-domain gate on the four unseen LTX scaffolds: guided samples reach 15.342 dB/0.6942 SSIM versus 13.777 dB/0.4160 for deterministic continuation, move contrast from 0.426 to 0.743 and edge energy from 0.486 to 0.830, retain 99.08% birth-cell adherence, and preserve a recognizable primary action in three of four classes. Correct and shuffled text remain tied once the true LTX video guide is present. The result therefore validates privileged scaffold-to-jewel transfer, not free prompt-only jewel generation. A new 0.88M-parameter scaffold/carry topology head then removes fitted birth cells, counts, and ranks for one held-out LTX continuation stride. Correct-scaffold slot F1 is 0.6636 versus 0.6092 shuffled, 0.6091 null, and 0.6222 train mean. When the frozen flow synthesizes every predicted rank, learned topology reaches 15.464 dB/0.6967 SSIM versus 15.492 dB/0.6997 with oracle topology, predicts 99.68% of the birth budget, averages 5,822 effective contributors/frame, and copies carried features with zero error. A compatible 1,024-rank flow then generates an empty-state initial stride and two continuations using only its own append-only field. Correcting a censored clip-start projection raises deterministic correct-scaffold PSNR from 12.408 to 14.570 dB and frame-zero visible density from 106–244 to 8,586–9,513, while stable IDs and carry remain exact. The selected field averages 7,493 effective contributors/frame and beats shuffled scaffolds by 4.116 dB. A foreground/motion-aware fine-tune reaches 14.972 dB/0.6610 SSIM but raises quiet-region temporal error by 3.8%; it is rejected under the predeclared stability gate. A two-stream control then integrates the selected base independently and copies lifecycle state exactly. Full-field appearance retains the fidelity gain but still flickers; restricting its residual to RGB in the top 20% scaffold-salient cells passes the four-class gate at 14.604 dB/0.6314 SSIM, lowers macro quiet error, improves foreground PSNR/edge and motion-boundary metrics in three classes, and keeps lifecycle coordinates, topology, density, and stable IDs bit-identical across 12 controls. A zero-initialized 74,067-parameter RGB adapter preserves the same exact ownership and improves its matched frozen base by 0.0098 dB while lowering macro quiet error, but captures only 29% of the full-flow PSNR gain. It therefore compresses part of the safe correction without passing the replacement gate or resolving the native-aspect motion-field noise. A zero-residual neighborhood set block then improves exact-count native PSNR by 0.332 dB, foreground PSNR by 0.761 dB, and all three macro error measures with the base frozen, but fails the three-class SSIM and visual-recognition criteria. This causally selects joint birth reasoning as the next direction without selecting the current checkpoint.

Gaussian reconstruction of video is not new: VeGaS, GSVC, GaussianVideo, GaRV, and related work predate this project. The research opportunity evaluated here is narrower: a corpus-level generative distribution that directly emits persistent, editable 2D-video space-time Gaussian fields. Because the field is a representation rather than a sampling objective, it can in principle be generated by flow matching, diffusion, or next-token prediction. Its hierarchy also suggests a structured multimodal event language in which text, persistent visual state, and edit operations share one sequence. We state these as provisional systems hypotheses, not priority claims.

1 Introduction

Pixel video models repeatedly synthesize dense frames and must learn temporal coherence through architecture, loss, or data. JEWELS asks whether coherence can instead be partly structural. We treat a video as a continuous three-dimensional signal over horizontal position, vertical position, and time. A moving visual element can then be represented by a Gaussian tube whose principal axis leans through time. The model is explicit: primitives can be selected, moved, recolored, carried between generation windows, culled at multiple levels of detail, and rendered at arbitrary times.

The intended system is not merely a codec. It is a text-to-video model whose native output is an editable space-time field. After generation, a viewer sees the video as a parallelepiped, drags a cursor through it, moves one or many jewels, and asks the model to resample only the vacated and destination neighborhoods. Untouched regions are hard constraints rather than soft conditioning.

The project has progressed through a sequence of falsifiable gates:

1. demonstrate that anisotropic space-time primitives can fit motion and ordinary video;
2. measure whether fitted sets are canonical enough for a shared generative model;
3. raise effective splat density without destroying temporal persistence;
4. learn a deterministic, count-aware local bottleneck for dense fields;
5. generate hierarchical latents with tractable attention;
6. prove exact edit locality and overlap-conditioned continuation; and
7. introduce text only after a leakage-safe prompt corpus exists.

Our contributions are therefore empirical and architectural rather than a completed generative benchmark. We report (i) an additive (u,v,t) Gaussian representation and gauge-free field features, (ii) density and tokenizer experiments that identify what preserves small moving subjects, (iii) controlled evidence that spatially aligned prefix context is essential for continuation, (iv) an edit/inpaint formulation with exact clean-region preservation, and (v) explicit negative results that bound what remains unproved. We further isolate deterministic conditional averaging as the current washout mechanism and show that a low-resolution semantic scaffold can recover global layout in stochastic jewel realization and transfer from UCF training to unseen prompt-generated LTX videos without mark-model retraining. We then replace fitted continuation topology with scaffold/carry-predicted counts and show that the frozen realizer remains within 0.028 dB of its oracle-topology control at the intended effective density. Finally, we remove fitted initial state, run three autonomous windows with exact append-only identity, isolate a lifecycle/appearance coupling that scalar foreground weighting does not resolve, and pass a conservative factorized control whose scaffold-gated RGB residual cannot alter lifecycle, topology, density, or identity. We then compress that path into a 74k-parameter base-locked adapter; exact ownership survives, but the student reproduces only part of the teacher's fidelity gain. A frozen-base neighborhood set block subsequently produces broad fidelity/error gains under exact base-owned topology, while failing the stricter structural/visual gate and selecting rendered set/trajectory supervision as the next bounded test. We additionally articulate two falsifiable systems hypotheses: that explicit persistent fields offer interaction properties unavailable from an opaque raster latent alone, and that a hierarchical field can serve as a multimodal visual event language.

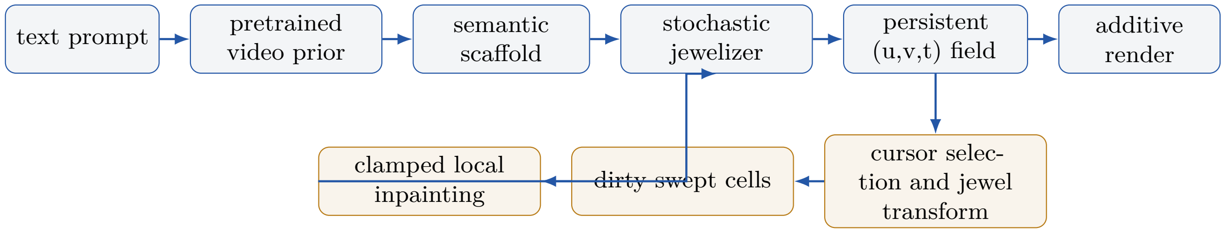

Figure 1: Revised target system. A pretrained raster video prior supplies global semantics; the jewelizer emits the persistent explicit state. Editing moves protected primitives, marks source, destination, and swept cells dirty, and resamples only those cells while the rest remains exact.

2 Relationship to Prior Work and Claim Scope

Modern Gaussian splatting was popularized for explicit 3D radiance fields by Kerbl et al. [8]. Dynamic-view methods subsequently added deformation, time, or native higher-dimensional primitives [10, 26]. Those systems reconstruct a view-dependent 3D scene; our target is the displayed 2D RGB video itself, without recovering physical depth or novel views.

That narrower video-representation lane is also established. Splatter a Video embeds monocular video into canonical 3D Gaussians with motion [21]. VeGaS uses folded Gaussian distributions to obtain time-conditioned 2D Gaussians and supports editing [19]. GSVC predicts frame-level 2D Gaussian sets from previous frames for compression [23]. GaussianVideo learns time-varying deformations of 2D Gaussians [9]. Most directly, GaRV encodes videos as 3D Gaussians in a space-time volume [18]. HyperGS amortizes reconstruction by predicting per-frame Gaussian representations feed-forward from input video [30]. Therefore, **we do not claim to be first to reconstruct 2D video with Gaussian splats, nor first to use a Gaussian space-time volume.**

Generative Gaussian work is substantial but usually targets 3D objects, rooms, or dynamic 3D scenes. GaussianCube and L3DG learn structured or latent generative models of 3D Gaussians [17, 27]. VideoRFSplat generates text-conditioned 3D scenes with a video model [4]; Diff4Splat predicts a deformable 3D Gaussian scene from an image, camera path, and optional text [15]. DiffSplat repurposes a large image diffusion prior to generate Gaussian splat grids and reports a rendering loss as important for coherence across novel views [11]. This supports inheriting mature raster priors and supervising rendered output rather than treating independent Gaussian-parameter losses as a complete appearance objective.

The jewel births are also a marked spatiotemporal point set. Point Set Diffusion directly models conditional spatial and spatiotemporal point processes without an intensity-function factorization [14], while point-cloud flow work emphasizes that permutation invariance and the coupling between noise and unordered targets materially affect the learned transport [7]. These results motivate a stochastic set objective; they do not remove the need for a coherent video-level condition in the present task.

Our provisional distinction is the combination of: (1) direct generation of the displayed 2D video as an anisotropic (u,v,t) Gaussian field, (2) persistent stable-ID carry and birth emission across overlapping time windows, and (3) hard-local jewel manipulation followed by constrained latent repair. Our August 2026 search found no published system with this exact combination. However, absence from a search is not proof of priority, and the current project has not yet shown generalizing text-to-video generation. We therefore use "toward" in the title and reserve any first-of-kind claim for a completed, externally benchmarked model and a broader review.

Table 1: Positioning against the closest video-Gaussian families. "Amortized" means a shared encoder predicts a representation from input video; it is not the same as unconditional or text-conditioned generation from noise.

Method	Native target	Temporal form	Shared prediction	Primary goal
VeGaS	2D video	folded 3D Gaussians, conditional slices	no	representation/editing
GSVC	2D video	linked per-frame 2D splats	no	compression
GaussianVideo	2D video	time-deformed 2D splats	no	representation; codec
GaRV	2D video	3D space-time Gaussians	encoder/decoder scheme	random-access decoding
HyperGS	2D video	predicted per-frame 2D splats	yes, from video	amortized encoding
Jewels (target)	2D video	persistent anisotropic (u,v,t) field	from noise and text	generation/editing

3 Method

3.1 Video as an additive space-time field

Let the normalized video domain be $\Omega = [-1, 1]^3$, with a query $\mathbf{x} = (u, v, t) \in \Omega$. Jewel i has center $\boldsymbol{\mu}_i \in \mathbb{R}^3$, symmetric positive definite covariance $\boldsymbol{\Sigma}_i \in \mathbb{R}^{3\times3}$, weight logit ℓ_i, base color $\mathbf{c}_i \in \mathbb{R}^3$, and a first-order color matrix $\mathbf{G}_i \in \mathbb{R}^{3\times3}$. Its contribution is

$$q_i(\mathbf{x}) = (\mathbf{x} - \boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i), \tag{1}$$

$$a_i(\mathbf{x}) = \sigma(\ell_i) \exp[-q_i(\mathbf{x})/2], \tag{2}$$

$$\widehat{\mathbf{I}}(\mathbf{x}) = \mathbf{b} + \sum_i a_i(\mathbf{x}) \left[\mathbf{c}_i + \mathbf{G}_i(\mathbf{x} - \boldsymbol{\mu}_i)\right], \tag{3}$$

where $\mathbf{b}$ is a learned constant background. This is additive accumulation, not ordered alpha compositing. The P1 color term avoids requiring many constant-color primitives to approximate smooth gradients.

Writing $\boldsymbol{\Sigma}_i = \mathbf{R}_i \operatorname{diag}(\mathbf{s}_i^2)\mathbf{R}_i^\top$ makes the motion interpretation explicit. If a principal axis of $\mathbf{R}_i$ tilts into time, the support is a sheared tube through the volume. Rendering a frame means querying the plane at its t coordinate. The representation therefore has continuous random access in space and time.

Training samples arbitrary voxels, minimizes RGB MSE, and periodically splits high-gradient primitives while pruning low-weight ones. Euclidean-center kNN supplies candidate splats; exact Mahalanobis weights are then evaluated for those candidates. Distance work is chunked to at most 100M point-center pairs, allowing 45k–120k-field experiments on an 8 GB RTX 2070 Super. Atomic checkpoints include optimizer and random-generator state, giving exact continuation across process failure and CPU/CUDA densification boundaries.

3.2 Gauge-free features and deterministic tokenization

Raw scale/quaternion parameters are unsuitable learning targets: quaternions double-cover rotation and covariance axes may permute. We instead use the unique log-Euclidean coordinate

$$\log \boldsymbol{\Sigma}_i = \mathbf{R}_i \operatorname{diag}(2 \log \mathbf{s}_i)\mathbf{R}_i^\top. \tag{4}$$

The upper triangle contributes six values to a 22-D feature: $[\boldsymbol{\mu}, \operatorname{triu}(\log \boldsymbol{\Sigma}), \mathbf{c}, \operatorname{vec}(\mathbf{G}), \ell]$. Decoding by eigendecomposition reconstructs a render-equivalent field with approximately 10^{-6} maximum feature round-trip error.

Dense fields are packed into a fixed 32^3 cell raster. The count-aware local encoder pools means, variances, counts, occupancy, and canonical rank features; the sparse decoder emits only the predicted variable number of jewels rather than allocating worst-case slots for every cell. A train-only PCA over non-overlapping 2^3 blocks maps eight 24-D fine cells into one 96-D coarse code. The resulting 16^3 grid is modeled with rotating axial attention: one block attends along u, the next v, and the next t. Three blocks communicate globally without materializing a 4096^2 attention matrix.

The full-field prior uses rectified flow [12]. For clean target latent z_1, noise z_0, and interpolation $z_s = (1 - s)z_0 + sz_1$, the model regresses the velocity $z_1 - z_0$ while receiving time and an optional CLIP condition [16]. Classifier-free dropout is required before the null branch is interpreted.

3.3 Persistent continuation

Long video is not generated by concatenating independent fixed clips. A stride retains jewels whose finite support intersects the overlap and copies their stable IDs and parameters exactly. The model predicts only new birth counts and 22-D birth marks in frontier-local coordinates. A $16 \times 16 \times 8$ raster of prefix counts and moments supplies spatially aligned context tokens. This factorization turns "splat density per unit time" into an explicit birth process: the field maintains a target contributor density while old jewels persist and new regions receive births.

3.4 Scaffold-conditioned birth topology

The first autonomous topology head receives two aligned $16 \times 16 \times 8$ fields. The scaffold field is trilinearly pooled RGB from the next 16 video frames. The carry field contains log density, occupancy, and mean opacity of jewels copied across the frontier. A 3D convolutional encoder with global pooled context predicts one occupancy logit a_j and one positive-count value $q_j = 1 + \mathrm{softplus}(r_j)$ for every cell j. Its continuous expected count is

$$\hat{n}_j = \sigma(a_j) q_j. \tag{5}$$

Training combines balanced occupied/empty binary cross-entropy, Smooth-L1 error on $\log(1 + q_j)$ in positive cells, total log-count error, and an L_1 error between normalized count distributions. The occupancy threshold is selected only on training fields by slot F1. Decoding rounds positive counts and expands each cell into nested ranks $0, \ldots, \hat{n}_j - 1$.

Topology uses a 1,024-rank contract because one initial UCF cell contains 919 births. The frozen continuation mark flow uses 512 ranks, so coupling validates every predicted continuation cell against that smaller capacity and raises an error rather than clipping. Carried features condition the topology head but are never transformed; generated marks are appended after the exact carry.

For empty-state generation we train a second, 1,024-rank mark flow over all 72 complete UCF views: 12 initial views receive an explicitly empty context raster and 60 later views receive only the preceding causal stride. At inference, each window recomputes context and carry from the generated field, expands the topology head's declared cell counts into ranks, samples every mark, converts frontier-local time to global time, and appends contiguous IDs. Existing rows are audited for bit-exact immutability after every append.

The observed beginning of a clip is a censored boundary, not necessarily a physical birth. A jewel assigned to initial time cell zero may therefore have finite support that began before frame zero. The projector leaves that lower boundary open only for the first local time cell of the first window; every later cell and continuation frontier remains strict. A matched control can restore the former strict initial projection without changing topology, noise, weights, or continuation semantics.

3.5 Stochastic mark realization and semantic guidance

The first prompt-conditioned continuation baseline predicts a sparse count raster and one canonical 22-D mark per occupied cell/rank using deterministic Smooth-L1 regression. Canonical rank provides a reproducible address, but it does not make the conditional output unimodal: several plausible future actors, poses, and foreground colors can share the same local prefix and prompt. A paired deterministic loss can therefore average mutually incompatible marks.

The controlled replacement holds target cells, counts, ranks, and carried jewels fixed and learns a rectified flow over only the marks. For data mark y_1, Gaussian noise y_0, and $y_s = (1 - s)y_0 + sy_1$, the network predicts $y_1 - y_0$ while receiving the aligned prefix raster, cell/rank address, flow time, and text. A hard projection maps sampled centers and finite-support birth starts back to their declared cells without moving valid exterior edge-cell centers.

An oracle-guide variant additionally cell-pools the true future stride rendered at 24×40. It tests whether a coherent video scaffold contains the missing information needed to realize jewels. Because that guide contains the answer at low resolution, it measures an architectural ceiling and cannot be interpreted as text-to-video inference.

The first multiscale control constructs three guide volumes: the available future video and low-pass versions at scales (2) and (4). A $2 \times 2 \times 2$ subgrid in every birth cell produces 24 local tokens, each containing RGB, horizontal/vertical/temporal RGB differences, local coordinates, and scale identity. A cell/rank/noisy-mark query cross-attends only to tokens in its addressed cell. For intermediate flow state y_s and predicted velocity v_θ, the denoised endpoint

$$\hat{y}_1 = y_s + (1-s)v_\theta(y_s, s, c) \tag{6}$$

is projected to the fixed topology and rendered on contiguous space-time patches. The tested visual objective combines RGB MSE, spatial/temporal edge L1, opponent-chroma L1, and patch SSIM while retaining feature-space flow loss for covariance stability. A follow-up hybrid keeps the entire cell-RGB 3D-convolution/global guide path and adds local tokens only as residual evidence.

3.6 A base-locked appearance adapter

The safe two-stream control can be compressed without granting the new module ownership of the field. Let v_0 be the frozen 2.13M-parameter mark-flow velocity and let a_ϕ be a compact adapter that receives the adapted mark state, v_0, addressed causal context and guide cells, cell/rank position, flow time, and text. For canonical RGB coordinates $\mathcal{C} = \{9, 10, 11\}$, scaffold-cell gate $g_j \in \{0, 1\}$, and calibrated strength λ, the adapted velocity is

$$v^{\text{app}}_{i,d} = \begin{cases} v_{0,i,d} + \lambda g_{j(i)} a_{\phi,i,d}, & d \in \mathcal{C}, \\ v_{0,i,d}, & d \notin \mathcal{C}. \end{cases} \tag{7}$$

The base trajectory is integrated independently from the shared Gaussian initial state. Every non-RGB value is copied from that trajectory after each Euler step, topology projection, and local-to-global transform; later topology, causal row selection, counts, density, and stable IDs also remain base-owned. The adapter has 74,067 parameters (3.48% of the base) and a zero-initialized three-channel head, so its initial sample is exactly the frozen sample. Training uses the same top 20% scaffold-saliency gate and a 288-by-192 coordinate grid that preserves the source's 3:2 aspect ratio. A teacher-distilled arm transfers only the validated full-flow RGB velocity difference; real-video render patches remain an independent target.

3.7 Neighborhood-coupled jewel birth sets

The independent mark flow already shares raw noisy-mark moments through a cell raster, but after rank, noise, and guide decoding it applies every residual block row-wise. We test whether this late independence causes structured speckle by adding one permutation-equivariant set block. For hidden rows h_i assigned to birth cell $j(i)$, the block scatters per-cell hidden mean, variance, log-count, and occupancy; a 3D convolution mixes those statistics across adjacent space-time cells; and the addressed neighborhood state is broadcast back to each row through a residual MLP. This costs $O(N + C)$ memory for N jewels and C cells, avoiding padded attention in initial cells that may contain hundreds of ranks.

The set block adds 414,016 parameters (+19.48%) to the 2.13M flow. Every historical tensor is loaded from the selected v1 checkpoint and frozen; only the new block trains. Its final projection is initialized to exact zero, making the augmented velocity bit-identical to v1 before the first update. Counts, rank expansion, Gaussian paths, hard topology projection, append-only carry, stable IDs, and rendering remain unchanged. Thus the experiment attributes any difference to learned local set coordination rather than a general model fine-tune.

3.8 Cursor edits and constrained repair

A cursor selects jewel centers or support intersections in the parallelepiped. Translating a group creates protected moved jewels and a dirty mask covering the source, destination, swept axis-aligned bounding box, and a cell halo. Let $m \in \{0, 1\}^K$ mark dirty latent cells and z^{clean} be the original hierarchy. After each Euler flow step, sampling reclamps

$$z_{k+1} \leftarrow (1-m) \odot z^{\text{clean}} + m \odot [z_k + \Delta s\, v_\theta(z_k, s_k, c, m)]\,. \tag{8}$$

The final merge contains untouched jewels, generated jewels decoded from dirty cells, and protected moved jewels. This guarantees locality mechanically; visual reconciliation at the destination must still be learned from edit tuples.

4 Experimental Protocol

Experiments use three tiers. A synthetic moving tube falsifies the basic motion representation. CUHK Avenue [13] supplies fixed-camera fitting and generation-mechanics experiments. UCF-101 [20] supplies diverse action and appearance tests. Unless stated otherwise, the short side is 160 pixels. The current prompt corpus uses 96-frame, 120k-jewel joint fits with temporal-preserving spatial splitting.

We intentionally distinguish metrics. Fit-training PSNR is often computed on sampled voxels and is not a publication-grade codec metric. "Full-volume PSNR" replays the complete saved field against the decoded source. Tokenizer PSNR compares continuous samples rendered from the fitted target and the decoded target, not decoded output against the source pixels. Energy distance evaluates latent sample distributions. Correct/shuffled/null continuation controls use identical targets and model weights. These quantities answer feasibility questions and should not be compared directly with published rate-distortion or FVD results.

5 Results and Discoveries

5.1 Reconstruction is viable, but not novel by itself

The synthetic anisotropic-tube test reaches about 55 dB. A real fixed-camera clip reaches 43.2 dB with 10k primitives; four recent UCF prompt-corpus fields reach 30.32–33.67 dB on exact checkpoint replay. Figure 2 shows the source, 120k-jewel reconstruction, and five-times-amplified absolute error for basketball. Class identity, player silhouettes, court geometry, and dominant colors survive. Error concentrates on thin limbs, the ball, faces, object edges, and motion boundaries.

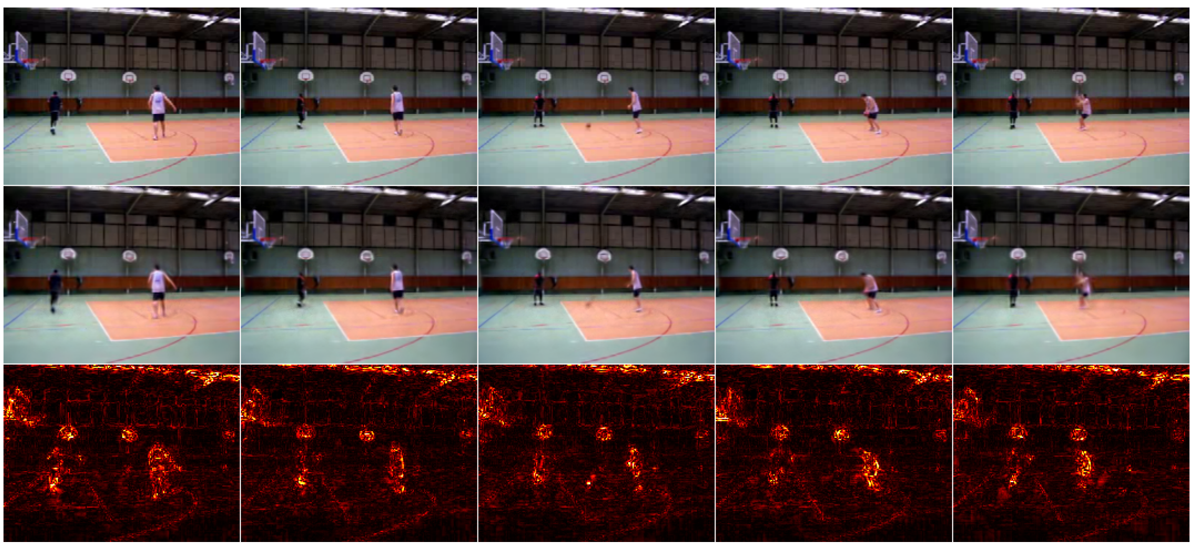

Figure 2: Exact 96-frame checkpoint replay sampled at frames 0, 24, 48, 72, and 95. Rows are source, 120k-jewel reconstruction, and 5× mean absolute RGB error. Full replay PSNR is 33.67 dB.

The covariance is not an ornamental parameter. Median anisotropy is about 2.8–3.3 on real footage, versus about 1.2 on the synthetic background-dominated test. More importantly, attempts to impose a soft Voronoi partition reduced both reconstruction and cross-seed canonicality. On one real clip, additive splats score 43.2 dB with a 0.621 normalized cross-seed Chamfer ratio; vanilla Voronoi scores 36.8 dB/0.745. A background pseudo-cell recovers some PSNR but not canonicality; Lloyd relaxation collapses to 33.0 dB/0.784. Equalizing cell sizes fights the desired content-adaptive allocation, so the branch was removed.

5.2 Useful density is a rate, not a raw count

At 45k total jewels, temporal 3σ support intersects roughly 6.2k splats per Avenue frame, but only 3.9k exceed 5% potential peak alpha and the opacity-weighted effective count is 3.3k. UCF is lower: 3.6k and 2.9k. Thus a nominally dense field remains below the desired 5k–10k effective contributors per frame.

The naive 90k isotropic control demonstrates why. It nearly doubles observed births per frame, shortens median support from 5 to 3 frames, increases effective contributors only 30.3%, and lowers PSNR. Figure 3 shows that extra count does not recover more useful detail.

Table 2: Matched UCF density control. The successful next control preserves temporal extent and splits spatial axes rather than shrinking every axis.

Measurement	45k isotropic	90k isotropic	Change
Peak alpha $\geq$ 5%/frame	3,595	4,724	+31.4%
Effective contributors/frame	2,933	3,822	+30.3%
Full-volume PSNR	34.136	33.062	−1.074 dB
Mean support lifespan (frames)	8.41	5.58	−33.6%
Median support lifespan (frames)	5	3	−40.0%
Observed births/frame	659	1,290	+95.6%
Fit runtime (minutes)	56.8	79.7	+40.3%

Figure 3: Source, 45k fit, and 90k isotropic fit at the same UCF frame. Doubling count increases short-lived fragments rather than useful detail and reduces full-volume PSNR.

5.3 A generative field prior works before it works well

An early permutation-invariant SetDiT flow model directly generated 6,471-jewel Avenue fields from noise. Scaling 5.8M, 58M, and 173M-parameter models under a frozen protocol monotonically improves CLIP-Frechet distance from 0.31 to 0.20 to 0.16 and training loss from 1.148 to 1.066 to 0.993 (Figures 4 and 5). Samples preserve scene layout, palette, chirality flips induced by mirror augmentation, and temporal stability, but fine pedestrians remain distorted. The flattening curve identified data and representation density, rather than further model scaling on 231 windows, as the next axis.

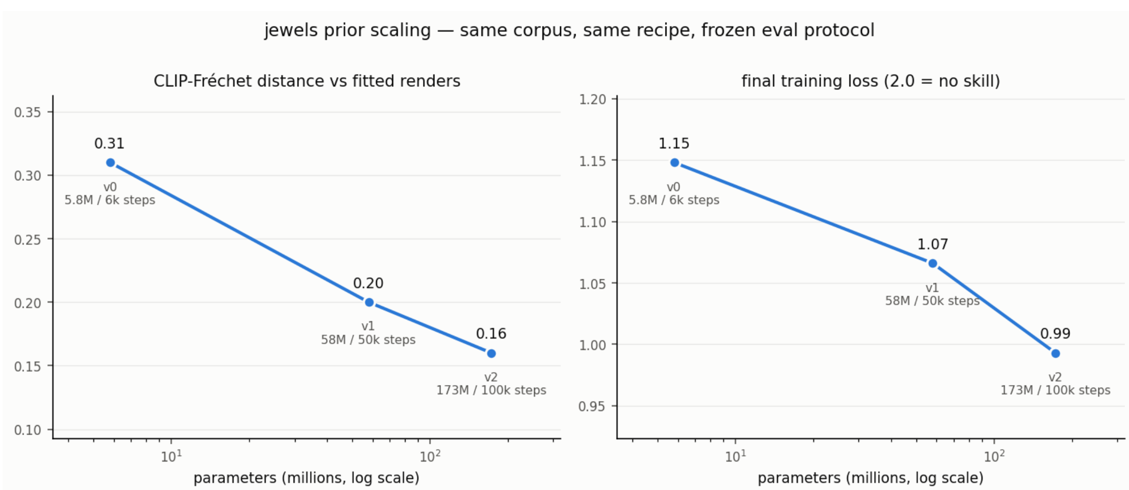

Figure 4: Frozen scaling protocol for the first direct field prior. Improvement is monotone but saturates on the small, single-scene corpus.

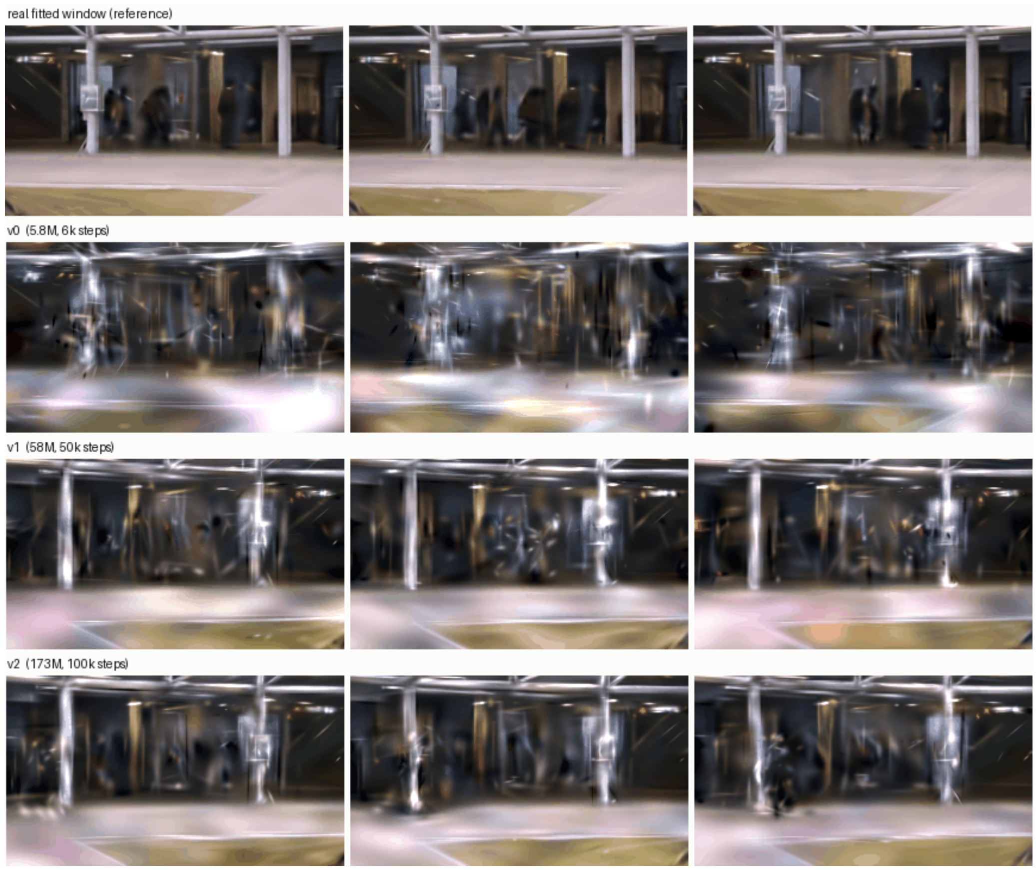

Figure 5: A fitted reference and samples from v0, v1, and v2 at three times. Larger models preserve the coarse scene more consistently, while people remain visibly under-resolved.

5.4 Locality is load-bearing for the tokenizer

On 121 dense Avenue windows, coarse cell pooling repeatedly achieved plausible aggregate PSNR while erasing people. The first source-held-out tokenizer to preserve recognizable pedestrian silhouettes uses a 32^3 grid, 24 values per cell, near-one-jewel average locality, and canonical-rank-aware decoding. Its 35-window audit reaches 19.987 dB and 99.14% count recovery. This remains a weak 1.26× numeric bottleneck. The 2^3 PCA hierarchy reduces 32,768 tokens to 4,096 and loses only 0.224 dB, making axial generation tractable.

Appearance is not solved. A bright red coat in held-out footage retains its figure but decodes tan. Rare foreground chroma is overwhelmed by uniformly weighted 22-D jewel losses and sparse uniform render samples. This is a regression-to-palette failure, not a channel-order bug. Foreground chroma and perceptual metrics must accompany aggregate RGB PSNR.

Cross-domain controls sharpen the conclusion. Frozen Avenue weights reach only 17.786 dB/86.06% count on UCF basketball; the identical architecture trained on that UCF window reaches 22.316 dB/97.13%. Equal-exposure joint training reaches 20.278 dB on held-out Avenue and 22.275 dB on UCF. The bottleneck can represent both, but absolute cell/domain specialization remains substantial.

Table 3: Selected tokenizer and hierarchy results. PSNR compares fitted target and tokenizer round-trip renders at fixed continuous samples.

Representation	Tokens	PSNR	Visual result
$12 \times 12 \times 6$, 128-D	864	18.940	people erased
$16 \times 16 \times 8$, rank-v2	2,048	20.428	people erased
$24 \times 24 \times 16$, pooled	9,216	19.730	faint traces
32^3, rank-aware, 24-D	32,768	**19.987**	people recognizable
16^3, PCA, 96-D	4,096	19.763	people retained

5.5 Generation, continuation, and editing separate into distinct gates

The 1.94M-parameter axial prior learns the coarse latent distribution: energy distance 0.2493 beats CLIP nearest-neighbor retrieval (0.3164) and repeated scene mean (0.8473). Yet correct CLIP conditioning is slightly worse than shuffled or unconditional conditioning (gain -0.00171). With only four training cameras and one image embedding per window, the model learns scene-family statistics, not semantic control.

Continuation gives a stronger causal control. Replacing one global prefix vector with aligned $16 \times 16 \times 8$ local tokens produces the results in Table 4 and Figure 6. Correct context reduces standardized birth-feature MSE by 62.7% against disjoint shuffled context and recovers 99.95% of births. Carried jewels are copied with zero error. This is still an overfit to four views of one clip, but it proves that spatial correspondence, not merely window identity, is required.

Table 4: Matched cell-local continuation control.

Metric	Correct	Shuffled	Null
Birth-feature MSE	**0.4475**	1.1988	2.1695
Birth-count MAE/cell	**0.3310**	3.7167	6.4994
Future-field PSNR	**19.870**	7.302	-29.735

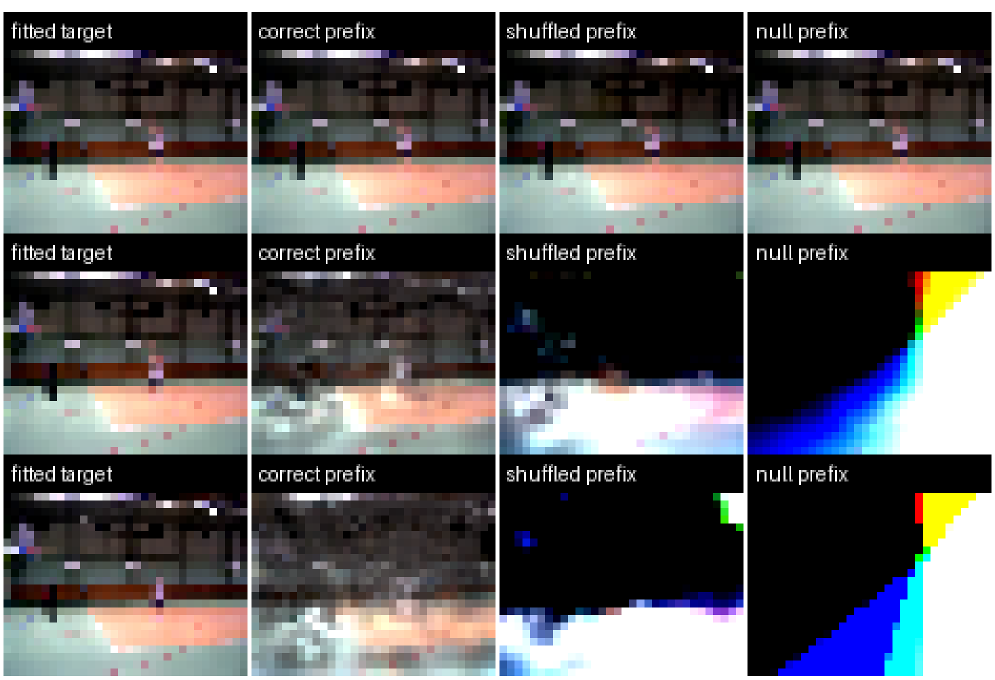

Figure 6: Start, middle, and end diagnostics for fitted target, correct prefix, disjoint shuffled prefix, and null prefix. The extreme null output is not a video baseline because this overfit was not trained with context dropout.

The editor translates 281 real fitted jewels, dirties 448 of 4,096 coarse cells (10.94%), and merges 45,244 final jewels. Every clean coarse and fine latent is bit-identical. The repaired strip, however, is visibly washed out. A mask-aware 5,000-step branch improves dirty latent MSE only from 1.8894 to 1.8848 and does not improve the visual. The result validates the editor invariant while falsifying the current repair learner.

5.6 Text preflight passes; direct deterministic generation washes out

The prompt smoke test selects Basketball, HorseRiding, PlayingGuitar, and ApplyEyeMakeup from four disjoint UCF source groups: 12 training videos and four group-held-out validation videos. Three training phrasings and one unseen evaluation phrasing per class are embedded with frozen OpenCLIP ViT-B/32. Each unseen phrase retrieves the correct training-phrase centroid; the minimum correct versus best-wrong cosine margin is 0.2395 (Table 5).

Table 5: Held-out text geometry before jewel-model training.

Class	Correct cosine	Best wrong	Margin
Basketball	0.8641	0.5881	0.2760
HorseRiding	0.9004	0.6610	0.2395
PlayingGuitar	0.9037	0.6546	0.2490
ApplyEyeMakeup	0.9102	0.6229	0.2873

All 16 dense 120k-jewel fits are complete. A 0.84M-parameter direct continuation model is trained on the 12 group-1–3 videos and evaluated on all four group-4 sources. With the prefix removed, the correct unseen text template has mark MSE 1.1579 versus 1.1794 for shuffled action text and changes free decoded counts by class. Thus prompt information reaches the model, but decoded videos remain washed out and do not depict reliable action geometry.

The proposed immediate remedy was to split the sparse topology head into occupied-cell classification and positive-count regression. We instead first held the entire target topology fixed. Table 6 shows that oracle cells, counts, and ranks improve mean PSNR by only 0.172 dB. Predicted geometry and predicted color/gradient each independently collapse the render, while predicted opacity is comparatively benign. Moreover, only 39.73% of predicted centers remain in their supplied spatial cell and 25.41% in the full space-time birth cell. This is a real contract defect, but it is not the main visual cause: even the oracle-topology render remains washed out.

Table 6: Held-out deterministic washout decomposition averaged over four classes. "Oracle topology" supplies target cells, counts, and ranks while predicting all 22-D marks. Other rows replace only the named target mark group.

Output	Mean PSNR	Finding
Free deterministic prediction	14.060 dB	Topology and marks predicted
Predicted marks, oracle topology	14.232 dB	Only +0.172 dB; still washed out
Predicted geometry only	14.434 dB	Geometry error destroys structure
Predicted color/gradient only	15.255 dB	Appearance regression independently washes out
Predicted opacity only	24.816 dB	Opacity is not the dominant failure

5.7 A stochastic jewelizer needs a coherent semantic scaffold

A 1.54M-parameter mark flow trained for 12,000 steps with oracle topology restores edge energy but not semantic coherence. Raw held-out samples average 15.064 dB and 84.4% of target edge energy, versus 14.232 dB and 59.8% for deterministic marks, yet the GIFs show structured speckle rather than recognizable action. Sampling avoids part of the conditional mean, but the local prefix and four-class text vector do not determine global scene layout.

A second 12,000-step control adds the true future only as a 24×40 guide. Table 7 and Figure 7 show the decisive result: the guided stochastic jewelizer reaches 16.555 dB and 90.5% edge energy, recovering the court/player, horse/rider, guitar/player, and face/hand macro-layout. A corrected hard projection maps generated marks to the declared topology while leaving every fitted target render-identical at 100 dB, so the gain is not a projection artifact. Residual high-frequency noise remains substantial. Correct and shuffled text are also similar under the oracle guide because the guide already supplies the future semantics.

Table 7: Held-out mark realization with target birth topology. Ratios are relative to the fitted target render; the stochastic row is raw because the original projection was not target-preserving, whereas the final guided row uses the corrected projection.

Mark generator	Mean PSNR	Contrast ratio	Edge ratio
Deterministic regression	14.232	0.574	0.598
Unguided stochastic flow	15.064	0.775	0.844
Oracle-video-guided flow	**16.555**	**0.856**	**0.905**

Figure 7: HorseRiding at three held-out future times. Columns compare the fitted target, carried prefix alone, projection control, deterministic marks, zero-guide flow, guided raw and projected flow, and shuffled text with the same oracle guide. The guide restores scene layout that topology and stochasticity alone do not.

5.8 Prompt scaffolds can be fitted at the intended contributor rate

The official distilled LTX-2.3 pipeline [5] generated a balanced set of 16 prompt videos: three training phrasings and one unseen evaluation phrasing for each of the four action classes. All 16 complete at 768×512, 49 frames, and 24 fps. Mean generation time is 172.98 seconds and mean aggregate GPU peak is 9,399 MiB on the RTX 4090. Independent contact-sheet review preserves the requested action and primary subject in every clip, so this corpus supplies substantially more coherent prompt semantics than the direct jewel model.

Before fitting all generated videos, we scale the established 120k/96-frame budget by space-time volume, which gives approximately 69k jewels, and round to a 72k ceiling. The first Basketball evaluation-prompt fit takes 1,523.1 seconds and replays at 28.182 dB. More importantly, its opacity-weighted effective density averages 6,384 contributors/frame and ranges 5,485–7,626; 7,389/frame exceed 5% potential peak alpha. Thus it lands inside the intended 5k–10k visual regime rather than merely increasing raw total count.

The frozen contract then transfers cleanly across all four actions (Table 8). Mean full-video PSNR is 31.472 dB; mean effective density is 5,966 contributors/frame and mean 5%-alpha count is 6,883/frame. PlayingGuitar is the instructive exception: its strict effective count is 4,597/frame, slightly below the nominal floor, despite a strong reconstruction. Density is therefore retained as a content-aware diagnostic rather than enforced as a blind scalar quota. ApplyEyeMakeup reaches 35.158 dB after a checkpoint-preserving 4090-to-2070S migration and preserves face, hand, eye motion, and palette. Figure 8 shows the weakest-PSNR Basketball case: court, players, ball motion, and red uniforms survive, while fine fence/tree detail and fast limbs remain soft. These pass as semantic-scaffold targets, not as final codec results.

Table 8: Frozen 49-frame/72k-jewel LTX reconstruction gate. Contributor counts are per-frame means; "effective" is opacity weighted.

Class	PSNR (dB)	Effective	Above 5% alpha
Basketball	28.182	6,384	7,389
HorseRiding	31.707	7,134	8,042
PlayingGuitar	30.839	4,597	5,339
ApplyEyeMakeup	35.158	5,749	6,762
Mean	31.472	5,966	6,883

Figure 8: LTX Basketball scaffold at frames 0, 12, 24, 36, and 48. Rows show decoded source, 72k-jewel reconstruction, and 5× absolute RGB error. Full replay PSNR is 28.182 dB; measured effective density is 6,384 jewels/frame.

5.9 Multiscale and render controls expose non-dominating trades

Table 9 reports the token-only multiscale model with render weight 2 against the cell-RGB v1 guide under identical topology, noise, sampling steps, held-out sources, and renderer. It raises target-relative contrast and edge energy, but lowers paired PSNR, mean frame-global SSIM, saturation, and temporal stability. Per-class behavior is inconsistent: ApplyEyeMakeup improves, Basketball is nearly flat, HorseRiding loses SSIM, and PlayingGuitar loses 0.948 dB. Visual inspection likewise finds cleaner local regions but no reliable action-level improvement. The branch therefore does not pass the realizer gate.

Two controls separate architecture from objective. Keeping the complete raster path and adding tokens with feature loss alone lowers every aggregate appearance metric, reaching 16.240 dB and 0.8345 SSIM. Keeping only the v1 raster architecture and applying render weight 2 raises contrast but loses 0.547 dB and 0.0087 SSIM. Reducing the predeclared weight to 0.5 improves mean SSIM by 0.0082, contrast ratio by 0.0795, and edge ratio by 0.0200, but loses 0.307 dB, reduces saturation, and increases excess temporal change. ApplyEyeMakeup improves by 0.584 dB, whereas PlayingGuitar loses 1.574 dB and overshoots both contrast and motion. The detail signal is real, but the present uniform tiny-patch estimator is not stable across actions.

Table 9: Matched four-class guided mark realization. Ratios target 1.0; temporal change above 1.0 indicates excess flicker/motion energy.

Guide/objective	PSNR	SSIM	Contrast	Edge	Saturation	Temporal
Raster, feature flow	**16.555**	0.8475	0.8555	0.9050	**0.9371**	**1.7236**
Tokens, feature+render (2.0)	16.419	0.8441	0.8694	0.9108	0.9194	1.7833
Raster+tokens, feature flow	16.240	0.8345	0.8519	0.8925	0.9209	1.7250
Raster, feature+render (2.0)	16.009	0.8388	0.8936	0.9021	0.8836	1.7725
Raster, feature+render (0.5)	16.248	**0.8557**	**0.9351**	**0.9251**	0.9167	1.8055

The token-only model exchanged cross-cell macro context for within-cell detail, and the residual token control shows that simply stacking both paths is not sufficient at this data scale. The render controls likewise reject scalar-weight tuning as the fix: their local losses can sharpen detail while amplifying motion inconsistently. We retain v1 for the cross-domain scaffold gate and reserve future render supervision for foreground/motion-aware sampling with an explicit temporal stability term.

5.10 UCF-trained jewel realization transfers to unseen LTX scaffolds

Before viewing any cross-domain output, we declare that v1 must beat deterministic UCF continuation in macro PSNR and SSIM, move contrast and edge ratios toward 1.0, retain at least 98% mean birth-cell adherence, and preserve a recognizable primary subject/action in at least three of four contact sheets. Training remains the original 12 UCF sources; validation replaces only the four group-4 clips with unseen LTX evaluation-prompt scaffolds. The first 16-frame continuation view, sampling noise, target topology, and renderer are matched across controls.

Table 10 shows that the gate passes. Guided flow gains 1.565 dB and 0.2782 SSIM over deterministic continuation. Contrast rises from 0.426 to 0.743, edge energy from 0.486 to 0.830, temporal change reaches 0.981 of target, and birth-cell adherence is 99.08%. Every class improves in PSNR and SSIM: PSNR gains span 0.590 dB to 2.627 dB. Projection without the video guide is insufficient, reaching only 12.810 dB; the semantic raster is the causal input.

Table 10: UCF-trained mark realization on four unseen LTX scaffolds. Appearance and temporal ratios target 1.0; birth-cell values report projected adherence.

Panel	PSNR	SSIM	Contrast	Edge	Temporal	Birth cell
Deterministic continuation	13.777	0.4160	0.4261	0.4860	0.7873	0.1021
Flow, no guide	12.810	0.3454	0.5048	0.6691	0.9583	0.9739
Flow, LTX guide	**15.342**	0.6942	**0.7428**	**0.8300**	**0.9815**	**0.9908**
Flow, guide + shuffled text	15.319	**0.6955**	0.7429	0.8260	0.9853	0.9904

Figure 9: One matched future time per LTX class. Columns show fitted target, deterministic UCF continuation, and guided projected flow; rows are Basketball, HorseRiding, PlayingGuitar, and ApplyEyeMakeup. Horse, guitar, and makeup retain recognizable primary subjects/actions. Basketball retains court/player layout but remains the weakest and blurriest transfer.

Correct and shuffled text are effectively tied under the true LTX guide: correct text changes PSNR by only +0.023 dB and SSIM by -0.00134. Thus the jewel model has not learned independent prompt semantics; those semantics arrive through the prompt-generated raster scaffold. This licenses the next experiment: predict occupied cells, counts, and ranks from that scaffold while stable-ID overlap state persists. Direct text-to-jewel distillation remains a later corpus-scale question.

5.11 Learned scaffold topology matches the oracle continuation ceiling

We train the 0.88M-parameter topology head for 3,000 updates on 72 complete initial and continuation views from the 12 UCF fields, then evaluate without adaptation on 12 views from the four LTX fits. The run takes 41.96 seconds on an RTX 2070 Super. Table 11 establishes causal scaffold use: the correct guide beats cross-class shuffled, null, and per-stride train-mean controls in slot F1 and count correlation. Removing carry drops slot F1 to 0.5374 and overpredicts births by 19.3%, making persistent overlap a load-bearing input rather than a merge detail.

Table 11: Held-out LTX topology over all initial and continuation views. Threshold calibration and the train-mean count field use UCF training data only.

Topology input	Slot F1	Count correlation	Birth ratio
Correct scaffold + carry	**0.6636**	**0.6010**	0.9390
Cross-class scaffold + carry	0.6092	0.4728	0.9341
Null scaffold + carry	0.6091	0.4628	0.8829
Per-stride UCF train mean	0.6222	0.3147	0.9378
Correct scaffold, no carry	0.5374	0.3440	1.1933

An oracle-mark rollout initially appeared to retain only 56–59% of fitted effective density: it discarded every predicted rank without an exact counterpart in one fitted decomposition. That is not a valid generative density test because additive Gaussian decompositions are non-unique. Two source-level cross-validation folds that add two LTX fits strengthen count discrimination but do not materially change exact-rank retention, arguing against fitting a larger corpus solely to fix that diagnostic.

The decisive experiment instead synthesizes every predicted rank with the frozen UCF mark flow. It evaluates the only LTX continuation view supported by the flow's 32-frame prefix contract, frames 32–48. The first 32 frames and carried jewels are fitted seed state; the learned topology uses correct scaffold RGB and carry. For the shuffled control only the topology guide is rotated to another class; the mark flow still sees the correct LTX guide and text. Thus any learned/shuffled difference is attributable to the topology field.

Table 12 shows that learned topology predicts 70,724 births versus 70,952 targets and all four class means lie between 5,598 and 6,449 effective contributors/frame. It loses only 0.028 dB and 0.0030 SSIM to generated marks under oracle topology. Its largest cell has 113 ranks, below the frozen flow's 512-rank capacity; spatial adherence is 100%, mean birth-cell adherence exceeds 98.8%, and carried features remain bit-identical with zero maximum error.

Table 12: Frozen mark realization on the held-out LTX 32–48 continuation. Appearance ratios target 1.0; density is opacity-weighted effective contributors/frame.

Panel	PSNR	SSIM	Contrast	Edge	Density
Carried only	13.529	0.3521	0.3263	0.3438	1,924
Oracle topology, generated marks	15.492	0.6997	0.7252	0.8197	6,252
Learned topology, generated marks	**15.464**	**0.6967**	**0.7202**	**0.8250**	**5,822**
Shuffled topology, correct mark guide	15.353	0.6927	0.7314	0.8170	5,781

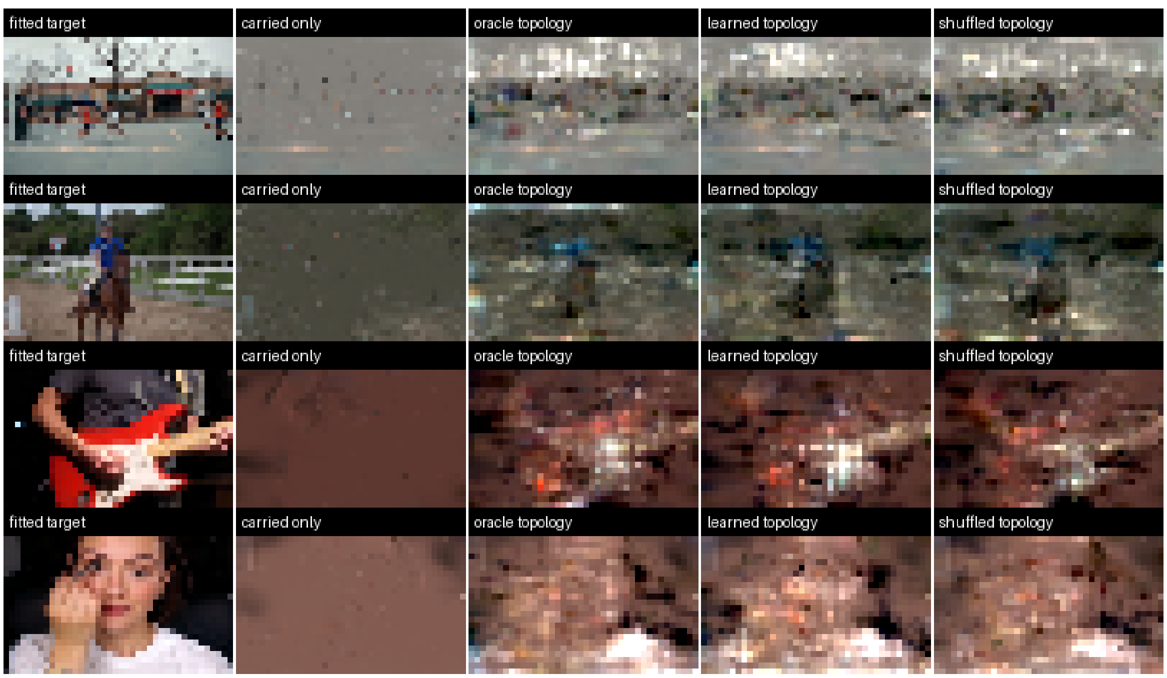

Figure 10: One middle continuation frame per LTX class. Columns show fitted target, carried-only state, generated marks with oracle topology, generated marks with learned topology, and generated marks with cross-class-shuffled topology. Learned and oracle topology preserve the same macro-layout and share the same motion-region noise, localizing that residual error in mark realization.

This passes learned topology, intended density, and exact carry for one held-out continuation stride. The learned/shuffled visual margin is modest because coarse occupancy is nearly dense and the correct mark guide supplies strong semantics, although slot overlap and correlation remain causally better. It licenses the fitted-seed-free rollout below.

5.12 Empty-state generation and two free-running continuations

One 2.13M-parameter universal mark flow is trained for 12,000 updates across the 72 UCF views in 162.14 seconds on an RTX 2070 Super. We freeze it and the topology head, then generate three 16-frame windows for each held-out LTX scaffold. Correct, cross-class shuffled, and null controls share text, weights, random seed, and a background derived only from the first scaffold stride. The initial window begins from zero jewels; later context, carry, and topology state come only from the control's own generated field. All results use deterministic PyTorch 2.12 CUDA kernels, 20 Euler steps, and seed 31.

The strict initial projection creates a mathematical opening ramp: its support-start constraint moves time-cell-zero centers to about three temporal standard deviations before frame zero, where a unit-opacity Gaussian contributes only $\exp(-4.5) \approx 1.1\%$. Treating frame zero as a censored observation boundary changes no learned parameter. Table 13 shows a 2.162 dB correct-scaffold gain, a larger correct-versus-shuffled margin, lower excess temporal change, and density closer to the fitted reference. Frame-zero jewels above 5% peak alpha rise from 106–244 to 8,586–9,513 across the four sources. Every prior and carried feature remains bit-identical, and every emitted ID is append-only and contiguous.

Table 13: Deterministic empty-state plus two-continuation rollout. Density is effective contributors/frame; seam is boundary change divided by ordinary temporal change.

Initial boundary	Correct PSNR	SSIM	Correct–shuffled	Density	Seam
Strict finite support	12.408	0.6041	3.601	10,403	1.035
Left-censored clip start	**14.570**	**0.6306**	**4.116**	**7,493**	**0.964**

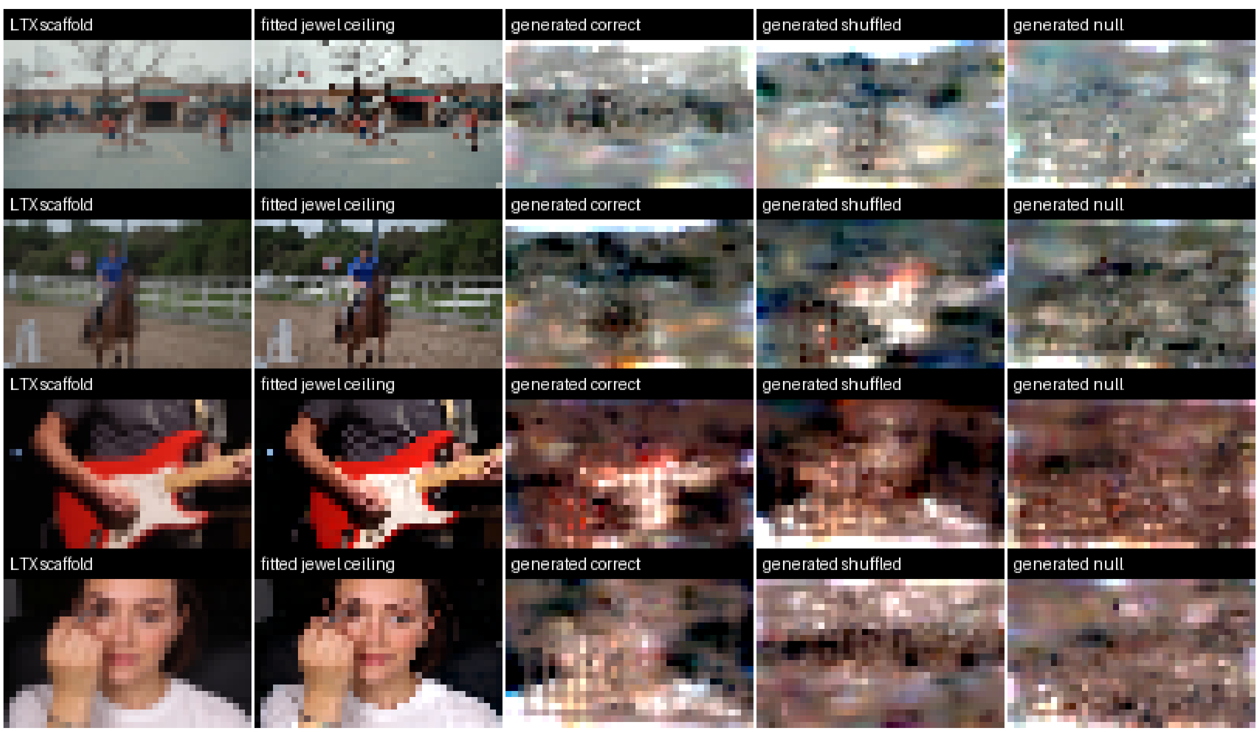

Figure 11: The first fully generated field at the first frame of the third window. Rows are the four LTX classes; columns show the LTX scaffold, fitted jewel ceiling, correct-scaffold generation, cross-class scaffold, and null scaffold. Correct guidance preserves action layout, but moving subjects remain noisy relative to the fitted ceiling.

This passes the structural generation gate: all 48 frames are jewels generated from a prompt-produced video scaffold and model-produced state, effective density lies in the requested 5k–10k regime, and seams are not worse than ordinary motion. It does not pass the final fidelity gate. Correct generation remains 7.27 dB below the fitted jewel ceiling, and the actor, hands, guitar, and face contain visible motion-region noise.

5.13 A trivial decode of the scaffold guide bounds every pixel-fidelity claim

The rollout's only pixel input per 16-frame stride is the $16 \times 16 \times 8$ cell-RGB guide computed from the resized LTX video. The obvious baseline is to decode that input with no learned model at all: invert the cell raster and trilinearly upsample each stride back to render resolution. Table 14 scores this decode under the identical deterministic rollout protocol, reference, and metrics.

Table 14: Guide-upsample baseline versus the generated field on the four held-out LTX scaffolds (48 frames, identical protocol). Ratios are candidate/target; 1.0 matches the target's energy.

Arm	PSNR	SSIM	Edge ratio	Temporal ratio	Fg. PSNR	Quiet MAE
Guide upsample (no model)	**21.081**	**0.9037**	0.515	0.508	**16.54**	**0.0064**
Generated correct (seed 31)	14.570	0.6306	**1.205**	**1.192**	10.23	0.0215
Fitted jewel ceiling	21.840	0.9310	1.449	1.887	—	—

The blurred upsample wins every reference-based fidelity metric in every class, by 5.5–7.8 dB PSNR, and sits only 0.76 dB below the fitted 72k ceiling. What it cannot do is carry detail: it retains roughly half the target's edge energy and temporal change, while the generated field restores both to approximately target level at target density. The same bound holds for the cel-shaded adaptation record (19.60 versus 15.40 dB). This is the perception–distortion tradeoff in an unusually clean form [2]: reference-based PSNR and SSIM [25] reward the blur and cannot demonstrate the generative stack's contribution. Two consequences follow. First, any future claim of rollout quality must either beat this baseline on fidelity or be scored where detail restoration counts, i.e. with perceptual and distribution metrics such as LPIPS [28] and FVD [22]. Second, the system's near-term value claim rests on what the baseline structurally lacks: an editable, persistent, object-decomposed field rather than pixels — and on closing the temporal-stability gap, since the trivial decode is currently 3.3× quieter in static regions than the generated field.

Scoring the same arms with per-frame AlexNet LPIPS at native 288×192 (Table 15) sharpens both consequences. The fitted 72k field reaches 0.098 LPIPS — seven times better than every other arm — so the representation's perceptual ceiling is nearly photoreal and the entire gap lives in generative realization. The perceptual ranking also splits where the pixel ranking did not: generated fields beat the blurred baseline on the two motion-heavy classes (Basketball 0.727 versus 0.783, HorseRiding 0.703 versus 0.792) and lose on the two stable close-up classes, localizing the remaining failure to generated temporal noise on smooth content rather than missing detail. The neighborhood-coupled variant improves LPIPS over the frozen base in three of four classes, independently confirming the coupling direction. Macro LPIPS still favors the baseline by 0.015, so no perceptual victory is claimed; the fitted ceiling's headroom identifies realization training — not the representation — as where that gap closes.

Table 15: Per-frame AlexNet LPIPS (lower is better) against the LTX target at native 288×192, 48 frames, deterministic seed-31 rollouts.

Arm	LPIPS	PSNR	SSIM
Fitted jewel ceiling	**0.0982**	27.521	0.9831
Guide upsample baseline	0.6761	19.347	0.8596
Generated coupled birth sets	0.6911	14.475	0.6031
Generated v1 base	0.6971	14.142	0.6001

5.14 Foreground weighting exposes lifecycle–appearance coupling

We initialize four 1,000-update arms from the passing universal flow. Target-derived scaffold saliency combines foreground/background deviation, rare chroma, spatial edge, and temporal change. Exploratory arms weight rendered salient patches, every 22-D feature, or only spatial/appearance dimensions; the final arm combines spatial/appearance weighting with a rendered quiet-change penalty. Temporal center and time-coupled covariance dimensions retain their ordinary feature loss but receive no extra saliency weight.

The decisive exact-seed comparison in Table 16 is not a null result. Global fidelity improves in all four classes; foreground-edge and motion-boundary error improve in three, and aggregate excess motion falls slightly. Yet quiet-region temporal error rises 3.8%, with PlayingGuitar and ApplyEyeMakeup each worsening by about 0.0018, while foreground PSNR falls in Basketball and PlayingGuitar. The branch therefore fails the predeclared consistency/stability gate and the base remains selected.

Table 16: Exact deterministic foreground/motion ablation. Lower is better for motion-boundary and quiet temporal MAE; temporal-change ratio targets 1.0.

Arm	PSNR	SSIM	Temporal ratio	Foreground PSNR	Motion MAE	Quiet MAE
Universal base	14.570	0.6306	1.1918	10.225	0.07235	**0.02145**
Spatial/appearance + stability	**14.972**	**0.6610**	**1.1872**	**10.456**	**0.07201**	0.02227

The causal conclusion is architectural. Excluding lifecycle dimensions from an extra loss does not freeze them: the shared flow trunk changes all velocities, and altered appearance state feeds back into later Euler steps and continuation rasters. The next control should integrate the frozen base trajectory independently, copy its temporal center/time-covariance state exactly, and learn a separate appearance/geometry residual stream. This turns lifecycle preservation from a soft penalty into an invariant.

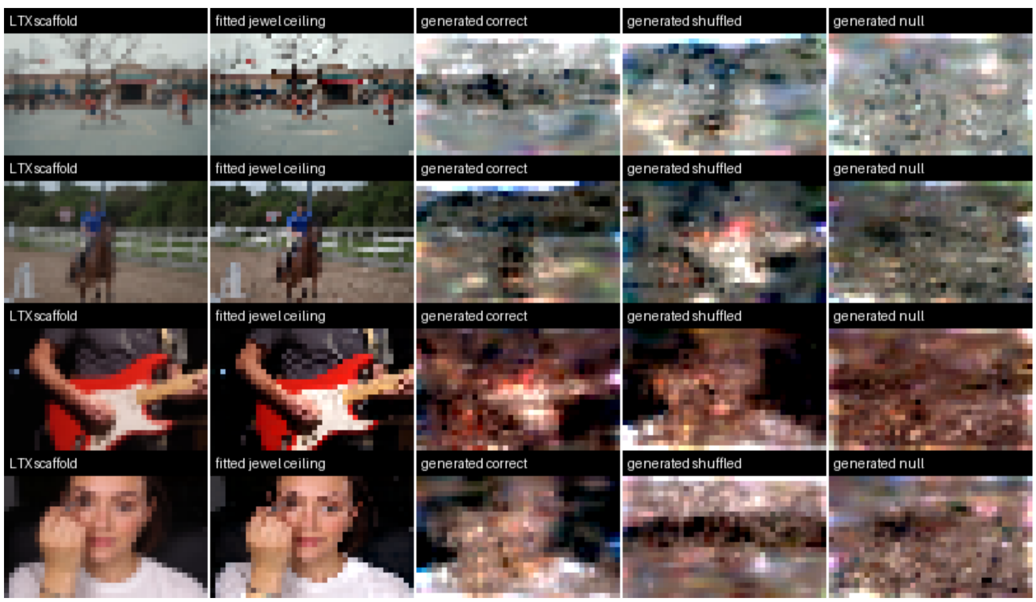

Figure 12: The rejected spatial/appearance-plus-stability arm at the first frame of the third window. Its aggregate fidelity gain is real, but the contact view does not show a uniformly cleaner subject, and the full temporal audit detects increased quiet-region flicker.

5.15 Exact lifecycle ownership admits a safe appearance residual

We next run the proposed two-stream control without training another checkpoint. Both streams share one Gaussian initial state, but the selected v1 flow is integrated independently and alone owns topology counts, causal row selection, stable IDs, temporal center, and time-coupled log-covariance coordinates. Candidate coordinates outside a declared residual set are copied from v1 after every Euler step, topology projection, and local-to-global covariance transform. Later topology is also predicted from v1 carry, so candidate opacity or geometry cannot change which jewels exist. The rejected saliency checkpoint supplies the second velocity field; its difference from v1 is the appearance residual. This reuse makes the experiment an architectural control rather than another optimization sweep.

Exact lifecycle copying does not by itself remove flicker. As Table 17 shows, allowing every spatial/appearance coordinate retains a 0.401 dB gain but raises quiet error 4.45%. Freezing opacity and RGB time-gradients barely changes the regression. Removing geometry cuts it to 0.32%, indicating that spatially displaced fixed-lifetime jewels still sweep through quiet pixels differently. The passing arm therefore allows only base RGB residuals in the top 20% of cells under the inference-available scaffold foreground/motion/chroma/boundary score.

Table 17: Deterministic lifecycle/appearance screens. The selected arm sacrifices most of the unconstrained gain to satisfy the predeclared quiet-stability invariant. Lower is better for the three MAE columns.

Residual	PSNR	SSIM	Foreground PSNR	Foreground edge	Motion boundary	Quiet temporal
Frozen v1 base	14.5700	0.63060	10.2252	0.68952	0.072349	0.021452
All spatial/appearance	**14.9713**	**0.65629**	**10.6827**	0.67613	**0.072201**	0.022406
Static spatial/detail	14.9148	0.65379	10.6600	**0.67587**	0.072205	0.022388
Color + spatial gradients	14.7476	0.63766	10.4657	0.68341	0.072240	0.021520
Color, all cells	14.7550	0.63781	10.4863	0.68263	0.072224	0.021516
Color, top 20% salient	14.6041	0.63140	10.3591	0.68486	0.072268	**0.021440**

The selected arm gains 0.034 dB PSNR and 0.0008 SSIM, adds 0.134 dB foreground PSNR, and lowers macro foreground-edge, motion-boundary, and quiet errors without changing the 7,493 effective contributors/frame. PSNR improves in all four classes; foreground PSNR, foreground-edge MAE, and motion-boundary MAE each improve in three. Across four sources and correct/shuffled/null controls, lifecycle coordinates, count tensors, topology, and stable IDs are bit-identical in all 12 rollouts. Correct guidance remains separated from shuffled by 4.141 dB. Two class-level quiet changes remain slightly positive (below 10^{-5}), so this is a conservative factorization proof, not a claim that motion fidelity is solved. A purpose-built zero-initialized adapter must now expand the safe residual beyond RGB without relaxing exact ownership.

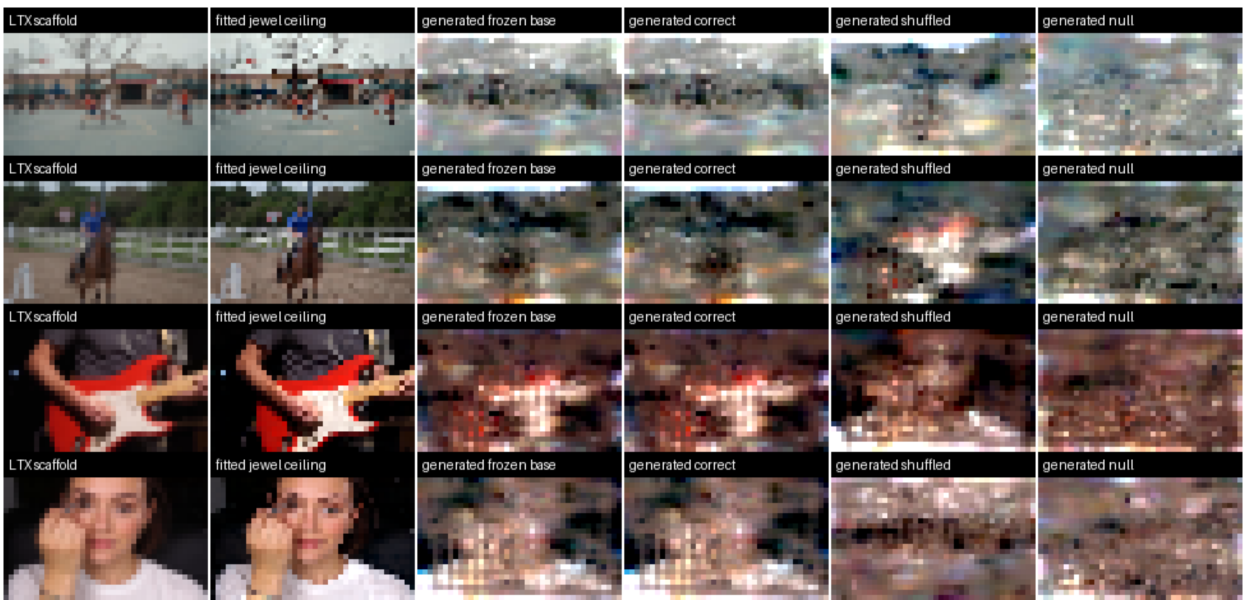

Figure 13: Selected scaffold-gated RGB residual at the first frame of the third window. The candidate is intentionally close to the frozen field: it passes stability by changing only base color in salient cells, while shuffled and null controls retain their expected semantic failure.

5.16 A compact adapter preserves state but only partly matches the teacher

We train the base-locked adapter on the same leakage-safe split. It has 74,067 parameters, versus 2.13M for either full flow, and trains for 3,000 updates in about four minutes on the RTX 4090. Direct target-velocity supervision improves held-out gated RGB velocity MSE by 1.53% at its best checkpoint, yet its 288-by-192 Basketball rollout loses 0.0079 dB and 0.00254 SSIM. This rejects single-time feature error as a selection metric. Distilling the validated full-flow correction instead produces a positive native-aspect result, but full strength raises Basketball quiet error by 1.87×10^{-5}. We therefore predeclare a one-half residual calibration for the final gate.

Table 18 reports the original 40-by-24 protocol only as an exact regression against Table 17; training and the primary visual audit use the native-aspect grid. The half-strength adapter improves its bit-identical frozen base by 0.0098 dB, raises foreground PSNR in three classes, lowers macro edge, motion, and quiet error, and keeps every class's quiet increase below 10^{-5}. Across all 12 correct/shuffled/null rollouts, every non-RGB coordinate, topology tensor, density, lifecycle value, and stable ID remains exact. Nevertheless, it captures only 29% of the full-flow

PSNR gain and 33% of its foreground-PSNR gain. Full strength captures 57% of the PSNR gain but narrowly violates the class-level quiet gate and still does not match the teacher. The compact replacement criterion therefore fails.

Table 18: Matched four-class adapter regression. The low-resolution grid is retained solely to compare with the previously selected full-flow control; lower is better for MAE columns.

Mechanism	PSNR	SSIM	Foreground PSNR	Foreground edge	Motion boundary	Quiet temporal
Frozen v1 base	14.5700	0.63060	10.2252	0.68952	0.072349	0.021452
Full-flow RGB teacher	**14.6041**	**0.63140**	**10.3591**	**0.68486**	**0.072268**	**0.021440**
74k adapter, $\lambda = 1$	14.5895	0.63085	10.3143	0.68647	0.072300	0.021441
74k adapter, $\lambda = 0.5$	14.5798	0.63074	10.2695	0.68799	0.072323	0.021446

The definitive native-aspect gate uses 288-by-192 grids and the stable per-source seeds 31–34. Relative to its independently integrated frozen base, the half-strength adapter raises macro PSNR from 14.1417 to 14.1481 dB and foreground PSNR from 9.9783 to 10.0077 dB. PSNR and foreground PSNR improve in all four classes; macro SSIM changes by $+2.69 \times 10^{-5}$, while foreground-edge, motion-boundary, and quiet temporal MAE fall by 9.29×10^{-5}, 1.02×10^{-5}, and 5.18×10^{-6}. The largest class-level quiet increase is 8.98×10^{-6}, inside the predeclared limit, and all non-RGB state remains exact. Thus the adapter passes its native frozen-base structural gate. It still fails the stronger replacement claim because the matched experiment above shows that it recovers only a minority of the selected teacher's improvement, and the native visual difference remains minute.

The failure is informative rather than destructive. The adapter proves that architectural state ownership survives a 28.7-times smaller trainable module. What it does not prove is that isolated RGB correction solves the visible noise: the native-aspect frozen and adapted fields remain nearly indistinguishable beside the fitted ceiling. A stronger student should be distilled on complete autonomous sampling trajectories, while the main realizer must couple neighboring births and their space-time geometry rather than treating color correction as the dominant missing capacity.

5.17 A coupled birth-set block produces the first broad autonomous gain

We train the single zero-residual set block for 3,000 updates on the same 12-source training split, using 288-by-192 scaffold preparation and leaving the 2.13M-parameter base frozen. The run takes 24.58 seconds on the RTX 4090. Step 2,250 minimizes the fixed-path held-out error at 1.53849 versus 1.54190 for the exact base (0.221% lower) and slightly increases correct-versus-shuffled scaffold separation. The error reduction is modest, so autonomous rendering remains the selection gate.

On the exact historical 40-by-24 protocol, the coupled model improves PSNR in all four classes and raises macro PSNR from 14.5700 to 14.9751 dB. Foreground PSNR rises from 10.2252 to 11.1027 dB; foreground-edge, motion-boundary, and quiet temporal MAE fall from 0.68952, 0.072349, and 0.021452 to 0.65655, 0.071758, and 0.020750. Foreground PSNR improves materially in three classes and is flat in PlayingGuitar. SSIM falls from 0.63060 to 0.62616 and improves only in Basketball, so this aspect-sheared regression is encouraging rather than sufficient. A one-half residual calibration reduces the useful gains and still does not restore class-wide SSIM, so the full block proceeds to the predeclared native-aspect gate.

At 288-by-192, the self-consistent autonomous comparison is broadly positive (Table 19). The coupled system gains 0.334 dB macro PSNR, 0.00300 SSIM, and 0.658 dB foreground PSNR while lowering foreground-edge, motion-boundary, and quiet temporal error. PSNR improves in all four classes, SSIM in three, foreground PSNR and edge error in all four, motion-boundary error in three, and quiet error in all four. ApplyEyeMakeup loses 0.00565 SSIM; Basketball alone raises motion-boundary error, by 0.000265. Effective density remains in the intended regime at 7,797 contributors/frame versus 7,886 for the base.

Table 19: Native 288-by-192 birth-set screens. "Self" lets topology react to each arm's carry; "exact count" gives the candidate the base's complete count/rank sequence. Lower is better for the three MAEs.

Mechanism	PSNR	SSIM	FG PSNR	FG edge	Motion	Quiet
Independent base	14.1417	0.60009	9.9783	0.25917	0.134010	0.023303
Coupled, self	**14.4753**	**0.60309**	10.6366	**0.25167**	**0.133451**	0.022668
Coupled, exact count	14.4742	0.59956	**10.7390**	0.25175	0.133469	**0.022582**

This comparison allows the frozen topology head to see each arm's own generated carry. Later feedback therefore changes the four-class birth total by 0.70%, even though the topology model, budget policy, seeds, prompts, and renderer are shared. We consequently run a stricter attribution in which the independent base owns every per-cell count/rank sequence and the coupled candidate receives fresh identically seeded noise. All 115 inherited tensors, all 264,870 births, and stable IDs are exact between arms. This paired result retains a 0.332 dB PSNR gain and raises foreground PSNR by 0.761 dB; foreground-edge, motion-boundary, and quiet error improve in all four classes. It also exposes the limit: SSIM improves only in Basketball and HorseRiding, and ApplyEyeMakeup foreground PSNR falls by 0.035 dB. Effective density remains inside the target regime but falls 2.03% because opacity, covariance, and lifetime marks change.

Contact review is less dramatic than the metrics: the horse/rider mass and several scene boundaries improve, but the paired output is not materially more recognizable in three classes. The strict selection gate therefore fails. The experiment validates neighborhood set state as a causal direction while rejecting this feature-loss-trained checkpoint as the finished realizer; the next test should supervise rendered set/trajectory structure directly.

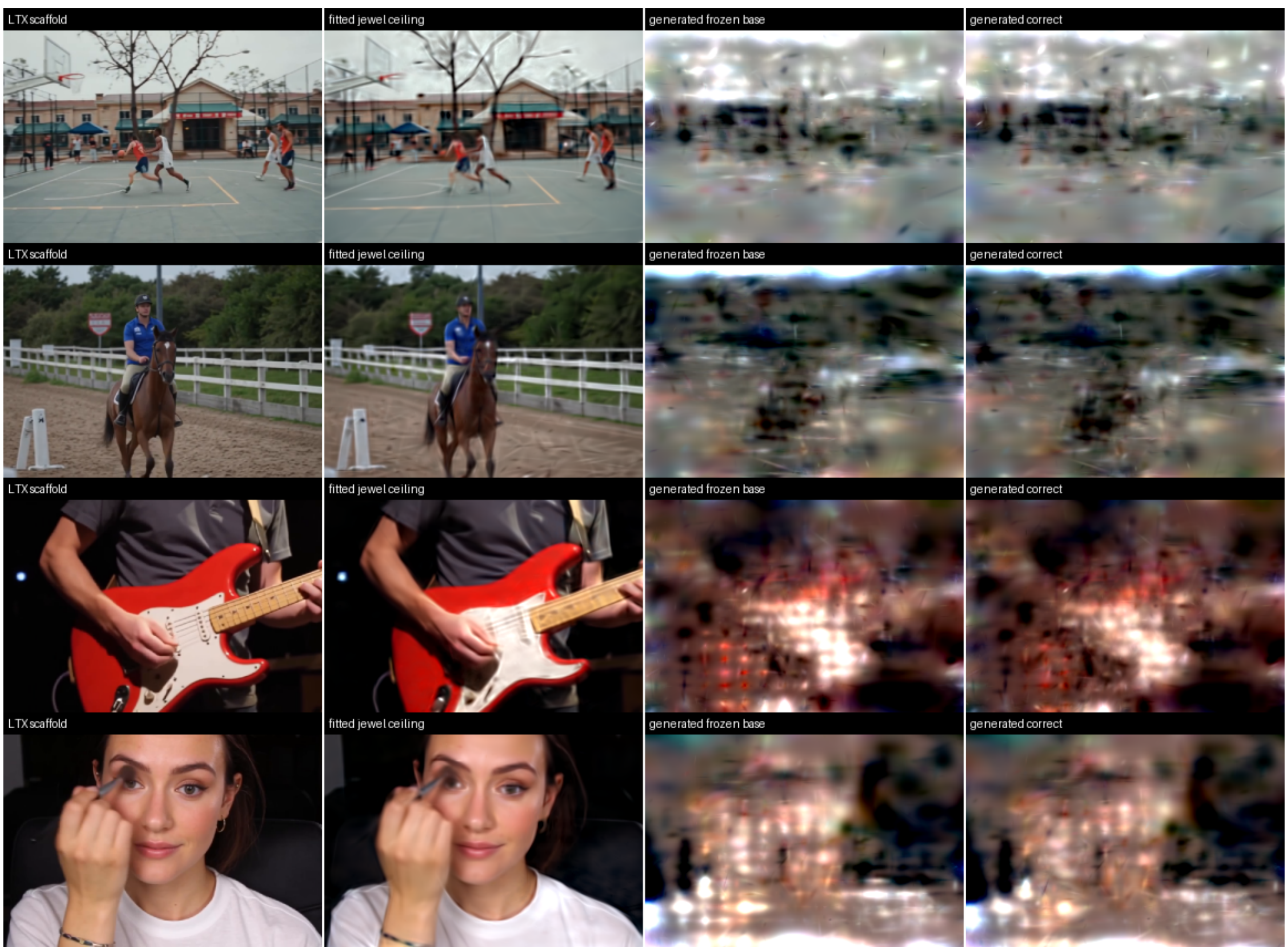

Figure 14: Exact-count native audit at the third-window frontier. The coupled field (right) shares every birth cell/rank and noise seed with the frozen base (center-right). Changes are real but subtle relative to the large gap to the fitted-field ceiling.

6 What Has Been Falsified

Failures have been used as architectural evidence rather than hidden tuning history:

Table 20: Selected falsifications and the resulting decision.

Failed hypothesis	Decision
Soft Voronoi improves canonicality	Removed; additive wins reconstruction and normalized cross-seed Chamfer.
More raw splats automatically increase useful density	Rejected; audit contributor rate and preserve temporal axes during split.
Aggregate PSNR selects a visually useful tokenizer	Rejected; require held-out motion silhouettes and foreground chroma checks.
Global prefix context is sufficient for births	Rejected; aligned cell-local context improves future-field PSNR by 3.41 dB.
The six-camera CLIP prior uses semantics	Rejected; correct condition does not beat shuffled/null.
Sparse count/topology error is the primary washout cause	Rejected; exact target topology adds only 0.172 dB.
Stochastic marks alone recover coherent video	Rejected; edge energy returns, but samples remain structured noise without a video guide.
Local multiscale tokens automatically improve the raster guide	Rejected; token-only and residual-hybrid controls both lose the matched visual gate.
Scalar render-loss tuning cures washout	Rejected; detail improves at weight 0.5, but one action becomes less faithful and less stable.
Scalar foreground weighting separates detail from lifecycle	Rejected; global fidelity rises, but quiet-region temporal error rises 3.8%.
Exact lifecycle copying alone removes quiet-region flicker	Rejected; full-field appearance still changes rendered motion, so the passing residual is scaffold-gated RGB only.
Single-time RGB distillation fully replaces the second flow	Rejected; the 74k adapter preserves exact state but captures only 29% of the teacher's gated PSNR gain at the passing strength.
Independent post-guide jewel rows are sufficient	Rejected; one frozen-base neighborhood set block improves exact-count native PSNR and all three error measures in all four classes.
Exact fitted cell/rank overlap measures generated density	Rejected; synthesizing all predicted ranks restores 5,822 effective contributors/frame and oracle-level rendering.
Mask visibility alone teaches repair	Rejected; train explicit edit tuples with protected-jewel summaries.
Further model scaling solves the sparse prior	Deprioritized; the curve saturates on 231 single-scene windows.

7 Limitations

- **No direct text-selective jewel generator.** The hybrid path now generates all 48 jewel frames from a prompt-produced LTX scaffold and its own state, but the jewel model's correct and shuffled text controls remain tied because semantics live in the raster scaffold.
- **Small and specialized corpora.** The generative results use one fixed-camera scene; the continuation result is an overfit; the prompt corpus contains only 16 videos.
- **Fixed camera and no cuts.** Global camera motion shears most primitives at once and spends density on camera motion. Camera tokens or explicit stabilization are future work.
- **Low resolution.** Current fits use a 160-pixel short side. PSNR values are not comparable to full-resolution codec papers without a matched rate-distortion protocol.
- **Conservative lifecycle factorization.** Exact two-stream ownership passes the macro stability gate only when the residual is restricted to RGB in the top 20% scaffold-salient cells. The dedicated 74k adapter preserves that ownership but only partly matches the full-flow correction; complete-trajectory distillation and broader residuals remain open.
- **Short autonomous horizon.** Fitted cells, ranks, marks, and carry are removed for all three generated windows, but the audit covers only 48 frames, four action classes, and one seed. Longer rollouts may amplify carry error

or exhaust the fixed topology/rank contract.

- **Weak rendered topology selectivity.** Correct scaffold topology beats shuffled/null in count space, but shuffled-topology rendering is only 0.111 dB below correct because occupancy is nearly dense and the mark flow still sees the correct guide. Finer/adaptive density fields may be needed for topology itself to control semantics.
- **Motion-region mark noise.** Learned and oracle topology share blur/noise on moving actors, hands, and thin objects. The coupled set block produces a broader 0.334 dB native gain, but reaches only 14.475 dB against a 27.521 dB fitted-field ceiling and remains visibly speckled. Cell mean/variance coupling is therefore useful but insufficient.
- **Pure-PyTorch rendering.** Exact kNN/Mahalanobis evaluation is useful for research but slower than a tile CUDA rasterizer. Current reconstruction is optimization-based, not amortized.
- **Edit quality.** Clean-region identity is exact, but dirty-region synthesis is not yet visually useful and may create additive collisions with protected moved jewels.
- **Reference-based fidelity metrics saturate below a trivial baseline.** A no-model trilinear decode of the scaffold guide beats every generated rollout on PSNR, SSIM, foreground PSNR, and quiet-region stability (Table 14); the generated field wins only detail- and motion-energy restoration. Until perceptual/distribution metrics (LPIPS, FVD) and external baselines are in place, no fidelity number in this report should be read as evidence that the generative stack outperforms trivial decoding of its own input.
- **Novelty is provisional.** Closely related literature is moving quickly. A formal claim requires a complete system, broader search, matched baselines, and external review.

8 Why an Explicit Jewel Field Rather Than a Raster Latent?

The relevant comparison is not "jewels versus diffusion." A jewel field specifies the state that is represented and rendered; diffusion, flow matching, and autoregression specify how a model learns or samples a distribution over states. Video diffusion models usually operate on dense pixel or learned spatiotemporal latent grids [1, 6]. The rectified-flow prior used here is itself closely related: flow matching supports Gaussian probability paths that include diffusion paths as special cases [12]. A future Jewels model could therefore use a diffusion or flow objective while retaining an explicit field as its decoded state.

Dense raster latents have formidable advantages: regular tensors map efficiently to accelerators, convolutions and attention are mature, topology changes are implicit, and large pretrained image and video priors already exist. The case for jewels is instead a set of interaction and state properties summarized in Table 21. These properties do not by themselves imply better photorealism or rate-distortion.

The oracle-guide experiment argues for using both representations where each is strongest. A pretrained raster video model can supply global scene semantics, pose, camera motion, and prompt knowledge; a stochastic jewelizer can convert that coarse scaffold into explicit persistent state. This is not a concession to pixel-only output: users still select, move, carry, render, and repair the final jewels. It is a bootstrap strategy that avoids asking a 12-video point-set learner to rediscover a video foundation model.

Table 21: Representation-level hypotheses relative to an opaque raster video latent.

Field property	Potential capability	Required evidence
Stable primitive identity	Exact carry across generation windows	Held-out autonomous rollouts without identity drift
Localized finite support	Hard-clamped local repair after direct manipulation	Perceptually convincing edits with unchanged clean regions
Adaptive primitive density	Spend capacity on motion, boundaries, and rare detail	Matched quality and rate at 5k–10k effective contributors/frame
Continuous (u,v,t) domain	Random-access time slices and resolution-independent rendering	Quality and speed against matched video decoders
Explicit marks and covariance	Select, move, recolor, split, or delete native state	User edits that do not require global video regeneration

The present motion noise is therefore a diagnostic, not something raw scaling is assumed to erase. The 45k-to-90k isotropic control increased effective contributors by only 30.3%, shortened support lifetimes, nearly doubled births, and reduced PSNR. More jewels should sharpen silhouettes only when densification splits spatial axes while preserving temporal support, and when motion-, edge-, and foreground-weighted samples prevent static background from dominating the objective. More fitting steps can let new primitives settle, while a shared generative prior may suppress inconsistent fit noise; neither can restore detail discarded by the representation or tokenizer. The direction should be rejected if matched temporal-preserving density fails to reduce motion-boundary error, if the tokenizer approaches one token per jewel, or if local edits still require near-global repair.

9 Jewel Fields as a Multimodal Event Language

Native multimodal generation is established prior art. Chameleon models arbitrary interleaved image and text token sequences [3]; Emu3 applies next-token prediction to discrete image, text, and video tokens [24]; and Transfusion combines language modeling and continuous image diffusion in one transformer [29]. We therefore do not claim that an LLM generating visual media is new. The narrower opportunity is to make the generated visual tokens persistent, spatially executable state rather than only codes that decode to a raster.

A jewel video admits an event factorization such as

$$\texttt{WINDOW},\quad \texttt{CARRY}(i),\quad \texttt{CELL}(c),\quad \texttt{BIRTH}(n),\quad \texttt{MARK}(\theta),\quad \texttt{MOVE},\ \texttt{SPLIT},\ \texttt{DELETE},\tag{9}$$

where i is a stable primitive identity, c is a space-time cell, n is a local birth count, and θ contains continuous position, covariance, opacity, and color marks. Text, audio events, camera instructions, selections, and edit commands could be interleaved with this stream. The model output would then be both media and an executable edit program: retained state is copied exactly, new events extend the frontier, and an edit dirties only the affected cells.

A flat sequence of 120k jewels per clip would be impractical and would discard spatial locality. The selected hierarchy suggests a tractable factorization: autoregress or mask-model the 16^3 coarse cell tokens and discrete event/count symbols, then use a local continuous head or conditional flow to emit fine 22-D jewel marks inside occupied cells. This also avoids pretending that every jewel is already an object-level semantic token; current primitives are low-level basis functions, and stable IDs are not automatically object identities. The multimodal-language hypothesis succeeds only if the hierarchy achieves substantial token compression, text controls held-out motion rather than retrieving a training clip, persistent events survive long rollout, and the same structured state supports cheaper local editing than full latent regeneration.

10 Path to a Promptable Model

The shortest evidence-driven path is:

1. freeze the passing two-stream control: v1 owns topology/lifecycle/IDs exactly, while RGB residuals are confined to the top 20% scaffold-salient cells;
2. retain the 74k zero-initialized adapter as a structural proof, but do not call it a full-flow replacement: at its passing strength it reproduces only part of the teacher correction;
3. if compact residual compression remains useful, distill complete autonomous teacher trajectories before expanding beyond RGB, and retain the class-level quiet gate;
4. address the larger visual bottleneck with a cell- or neighborhood-coupled set realizer whose births share local space-time geometry rather than sampling every addressed rank independently;
5. extend the autonomous audit beyond three windows and across multiple prompts/seeds before interpreting exact short-horizon carry as long-video stability;
6. evaluate finer or adaptive topology only if correct-versus-shuffled rendering remains tied after the mark guide is controlled;
7. train edit tuples with moved protected jewels, dirty masks, and local scaffold crops so source/destination repair is learned while clean regions remain exact;
8. build a 1k–10k captioned jewel corpus and distill the scaffold pathway toward direct text-to-jewel generation after the coupled realizer passes prompt and rollout gates.

A publication-grade comparison should include GaRV/VeGaS/GSVC-family reconstruction controls, pixel-space video baselines, parameter count, fitting/encoding time, memory, random-access render speed, rate-distortion, FVD or VBench-style generation metrics, edit locality, and prompt selectivity. The unique value proposition is not winning every codec metric; it is combining continuous temporal structure, persistent generation, direct manipulation, and constrained repair in one native representation. A longer-term foundation-model path would expose the coarse space-time hierarchy and event vocabulary to a multimodal transformer while retaining a continuous local generator for fine jewel marks.

11 Conclusion

The broad idea that Gaussian splats can reconstruct video is established prior art. The present work contributes a disciplined attempt to turn that representation into a generative and editable medium. The strongest findings are structural: anisotropic space-time support is genuinely used; effective density depends on temporal lifetime rather than raw count; near-one-jewel tokenizer locality preserves small actors; spatially aligned prefix context is essential for births; and local editing can preserve untouched latents exactly. The new washout audit adds a crucial causal result: correcting topology does not correct appearance, stochastic marks restore detail without semantics, and even a tiny coherent video scaffold substantially restores layout. The cross-domain gate strengthens that last result: without retraining, the UCF jewelizer improves all four unseen LTX classes over deterministic continuation and preserves three recognizable actions.

The project is therefore past the representation-mechanics question but before the text-to-video claim. It is also past the one-stride target-topology question: learned scaffold/carry counts feed the frozen mark flow at the intended density and reach the oracle-topology ceiling. It is now also past the fitted-seed-free short-rollout question: an empty initial field plus two continuations preserve append-only identity at the intended density, and a matched boundary control explains the former opening washout without retraining. The next milestone is not another scalar loss sweep. Foreground/motion weighting yields real detail but trades it against quiet-region stability, so the two-stream control separates an exact frozen lifecycle trajectory from appearance. It passes only under a conservative scaffold-gated RGB residual: topology, density, lifecycle, and stable IDs stay exact, three-class subject metrics improve, and macro quiet error falls, but the gain is just 0.034 dB. The zero-initialized compact adapter now preserves that ownership with 3.48% as many parameters as the base, but at its passing strength recovers only 0.010 dB and 29% of the teacher's gain. A first neighborhood-coupled set block now validates the proposed direction: with the base frozen and exact base-owned counts it gains 0.332 dB at native aspect, improves foreground PSNR by 0.761 dB, and lowers all three macro error measures. Yet it misses the three-class SSIM criterion, the contact-sheet change remains subtle, and the fitted-field gap remains 13.05 dB. The next realizer step should therefore train richer set geometry or trajectory-level rendered structure, not merely extend the same row-wise feature loss; complete-trajectory adapter distillation is a secondary compression problem. The hybrid preserves the distinctive explicit state while inheriting semantic competence that the current corpus cannot teach; prompt semantics still live in the pretrained scaffold, and direct distillation remains a later gate.

A Reproducibility Snapshot

Table 22: Current reproducibility-critical constants.

Component	Frozen or selected setting
Coordinate domain	normalized $[-1, 1]^3$ in horizontal, vertical, and time
Primitive feature	22-D: center, log-covariance, P0 RGB, P1 color gradient, weight logit
Renderer	additive P1, learned background, kNN 64, 100M distance-pair cap
Dense prompt fit	96 frames, 160-pixel short side, 16k init, 120k maximum, spatial split
Fine tokenizer	32^3 cells, 24-D local latent, sparse variable-count decode
Hierarchy	non-overlapping 2^3 train-only PCA, 96 retained components, 16^3 codes
Prior	rectified flow, rotating axial attention, nullable CLIP/text condition
Continuation	empty initial state plus two 16-frame strides; generated-state context/carry; append-only IDs
Topology generator	0.88M 3D cell-RGB/carry head; 16^3 cells, 1,024-rank initial contract
Mark generator	selected 2.13M 1,024-rank 22-D rectified flow; 20 deterministic Euler steps
Appearance control	independent frozen v1 trajectory; RGB residual in top 20% scaffold-salient cells
Compact adapter	74,067 parameters; RGB only; $\lambda = 0.5$ passing quiet calibration; exact base ownership
Semantic control	prompt-generated LTX 24×40 guide; three-window autonomous rollout passes structurally
Editing	swept source/destination dirty mask, protected moved jewels, exact clean-cell clamp

B Novelty Checklist Before Submission

1. Update the literature search through the submission date, including citations that cite VeGaS, GaRV, HyperGS, and Diff4Splat.
2. Replace the provisional positioning paragraph with a feature-by-feature comparison verified against code or supplements, not abstracts alone.
3. Demonstrate text-conditioned generation from noise on held-out prompts and sources.
4. Extend the passing initial-plus-two-continuation result across longer horizons and multiple seeds/prompts; replace one-step adapter distillation with trajectory supervision or document the compact branch as a structural control only.
5. Demonstrate a moved-object repair in which clean regions are exact and the dirty region is perceptually plausible under multiple samples.
6. Release the protocol, checkpoints, videos, and a matched baseline script.

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